Workshop #1

Aeroservoelasticity of Fixed and Rotary Wing Aircraft

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Abstract

This report presents the aeroelastic analysis of the "Goland Wing". Sensitivity to meshing, convergence of modal frequencies, and flutter speed under various structural and mass modifications are discussed, following the guidelines of the 2025/26 course.

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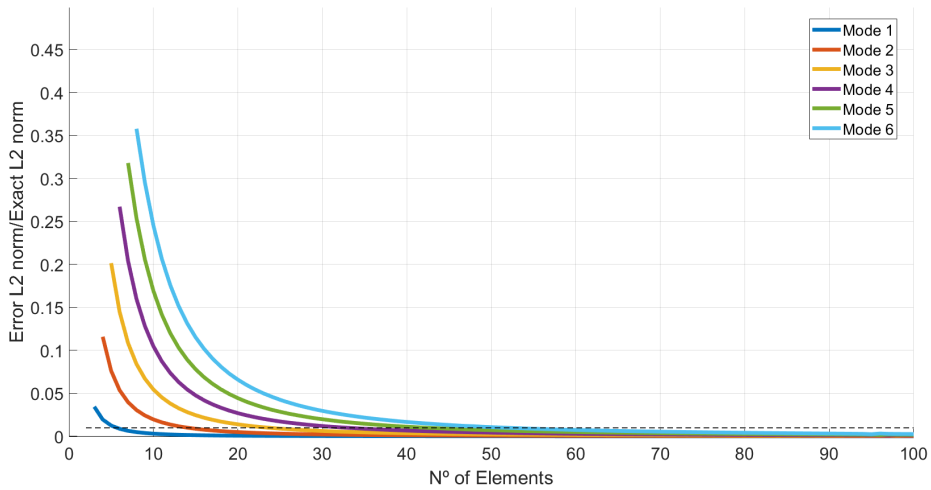
1 Sensitivity to the number of elements

Objective: Verify the minimum number of elements in the FEM model to keep the error in the modal shape approximation below 1%

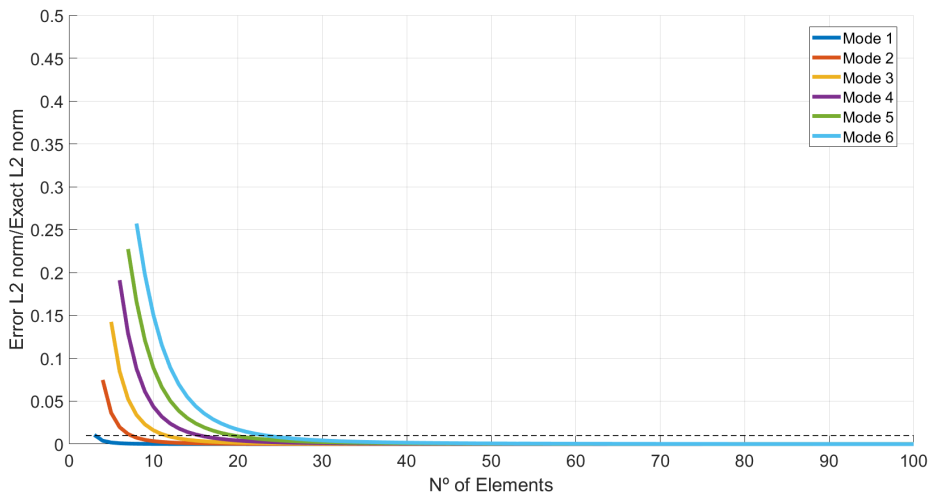
In this section, the convergence of the finite element model is analyzed considering the Goland Wing beam model. For this matter, the provided codes `ExactBeamBendingModes` and `FEM_beam_bending` have been used.

For each number of elements used, FEM modal shapes have been interpolated using **MATLAB** function *gridedInterpolant*, and then the resulting function evaluated in N_{eval} points. Both linear and cubic interpolations have been used and the convergence compared, to see how much improvement of accuracy can be obtained.

The modal shape error has been computed as the **error L2 norm** divided by the **exact L2 norm**, both evaluated in the N_{eval} points.

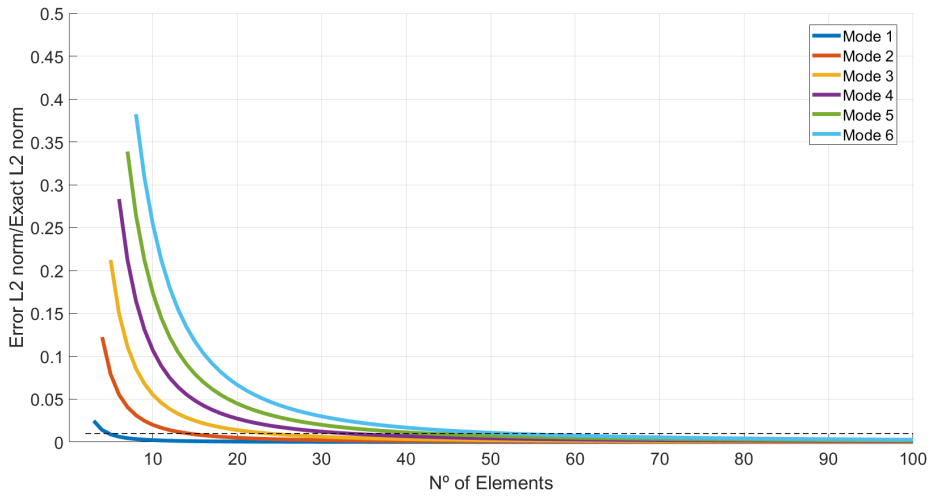


(a) Linear interpolation

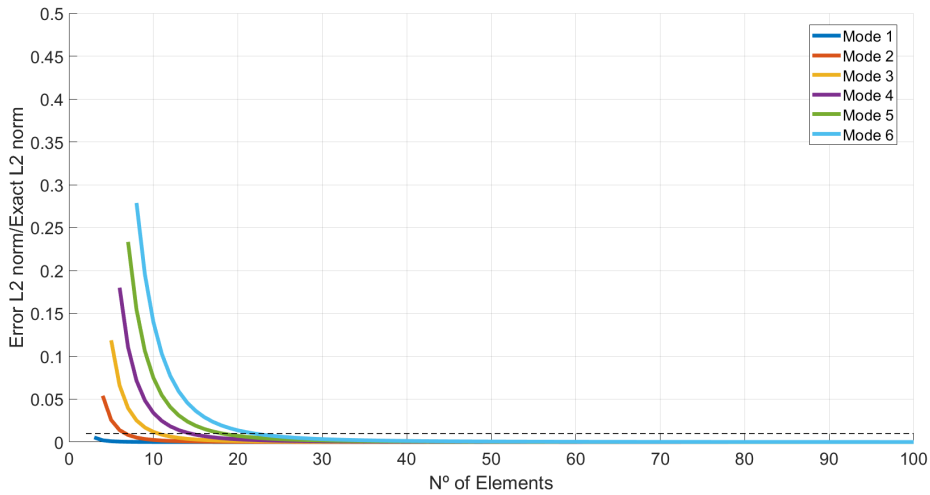


(b) Cubic interpolation

Figure 1: Convergence of modal shape approximation error with number of elements for the 6 first bending modes



(a) Linear interpolation



(b) Cubic interpolation

Figure 2: Convergence of modal shape approximation error with number of elements for the 6 first torsion modes

The results show how the interpolation used affects the convergence, reducing to half the elements needed with cubic. Additionally, both torsion and bending need a similar discretization to capture correctly the first 6 modes.

2 Convergence of Modal Frequencies

Objective: Analyze the number of uncoupled modes needed to capture correctly the frequency of the first 6 coupled ones

In this section, the composition of the first 6 coupled modes of the Goland Wing will be analyzed indirectly, by measuring the error of the frequencies obtained.

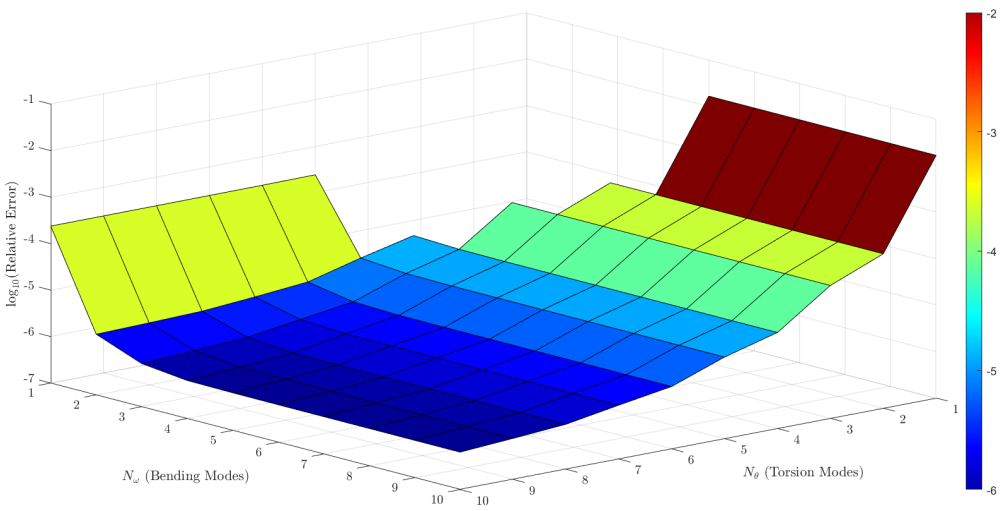
For this section the code `Goland_coupled_using_modal` has been used, modifying the bending and torsion uncoupled modes used to compute the proper orthogonal, by addition of the out of diagonal coupling terms in the inertia matrix.

To compute the error, the Richardson approximation for the 6 frequencies has been used:

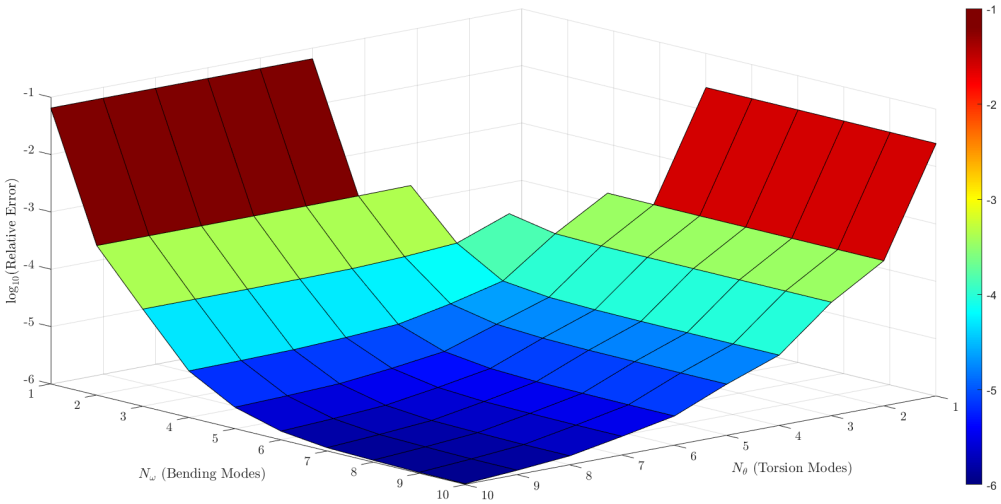
$$\omega_{exact} \approx \omega(N) + \frac{\omega(N) - \omega(2N)}{2^p - 1} \tag{1}$$

With p equal to 2 and N equal to 6 modes, doing first the bending modes approximation and consecutively the torsion modes.

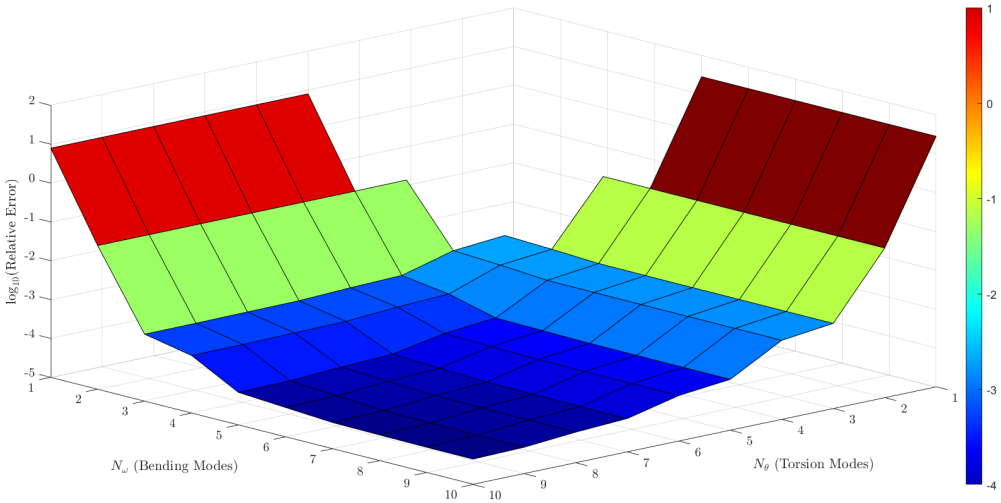
Below, all 6 graphics (one for each proper orthogonal mode) are plotted, using the logarithm of the error to make the convergence of the error more clearly visible.



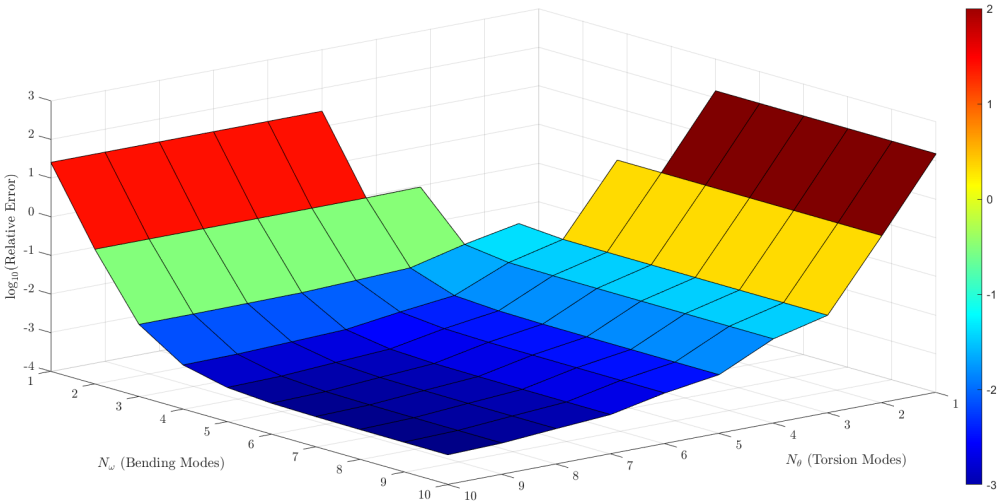
(a) Mode 1



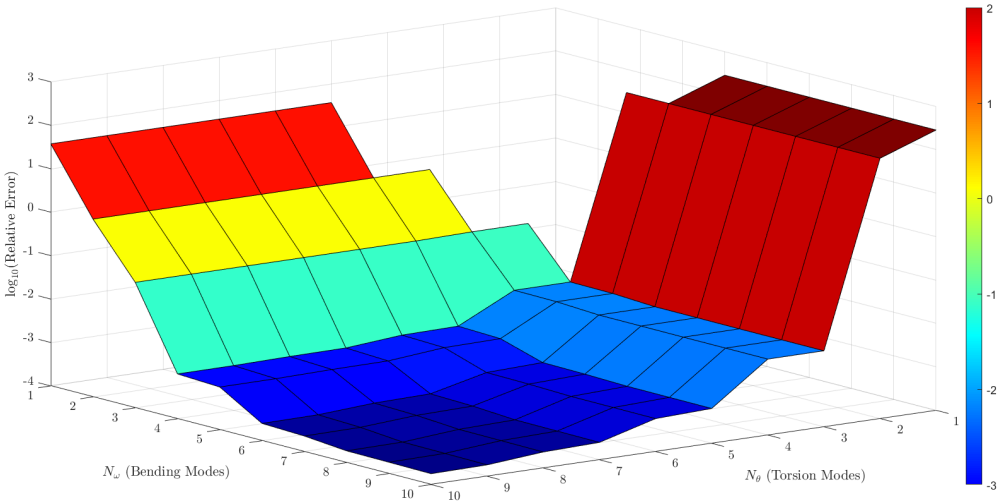
(b) Mode 2



(c) Mode 3

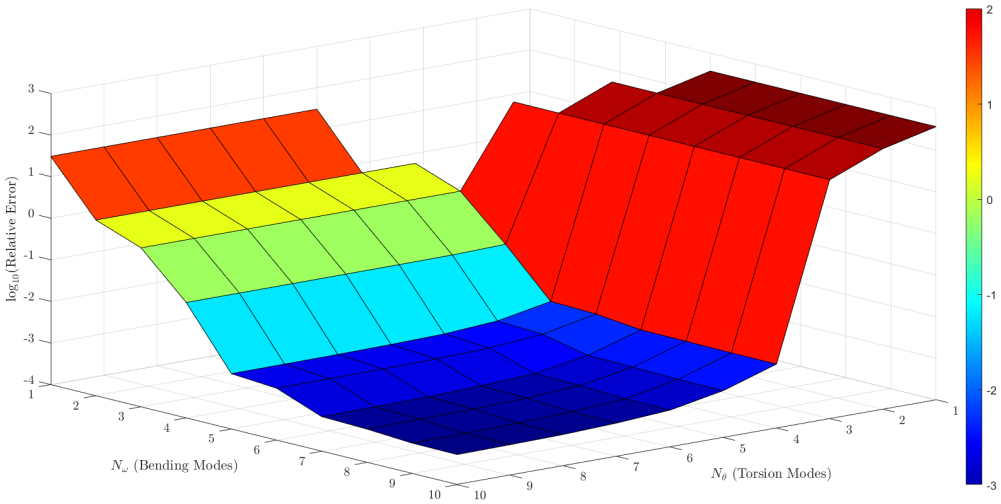


(d) Mode 4



(e) Mode 5

Figure 3: Convergence of modal frequencies (continued).



(f) Mode 6

Figure 3: Convergence of modal frequencies error (N_w vs N_θ)

If a convergence of log10(error) equal to 0 is considered good enough, the mode 6 is in this case the one that requires the most both bending and torsion uncoupled modes (3 bending and 4 torsion). For the first 2 modes, it has been considered a minimum of one uncoupled for each, so it cant be seen clearly in the plots and the modal shape will need to be checked. Following, a table displaying the approximated number of bending and torsion modes needed for each proper ortoghonal mode:

Table 1: Description of the first 6 coupled modes in terms of bending and torsion components.

Coupled Mode #	**Bending Mode Component**	**Torsion Mode Component**
1	1st Bending	-
2	1st Bending	1st Torsion
3	2nd Bending	2nd Torsion
4	2nd Bending	3rd Torsion
5	2nd Bending	3rd Torsion
6	3rd Bending	4th Torsion

This table is just the estimated most influential uncoupled modes, so it can be seen for example that in 4th and 5th they are both the same, but in 4th the bending has more impact on the final mode while in the 5th its the torsion. This analysis is important in order to reduce the computational effort, because only the coupled modes that are dangerous in the flutter phenomena need to be kept in order to capture it correctly, and so you also fix the number of uncoupled modes.

3 Optimal Modal Basis

Objective: Estimate what is the minimal set of proper orthogonal modes that should be used to reduce the error in the flutter speed estimate below 1%

In this section, continuing with the final conclusion of last one, the main mode inducer of the flutter phenomena in the Goland Wing will be detected, by analyzing the flutter speed obtained taking into account a different number or proper orthogonal modes. For this matter, the code `Goland_Flutter_w_Modes` has been used, modifying it for the N coupled modes considered to be an input. It has also been included an interpolation of the eigenvalues to obtain with more precision the flutter speed, and will be used in all the following tasks.

Below, the flutter speed obtained by different number of coupled modes involved:

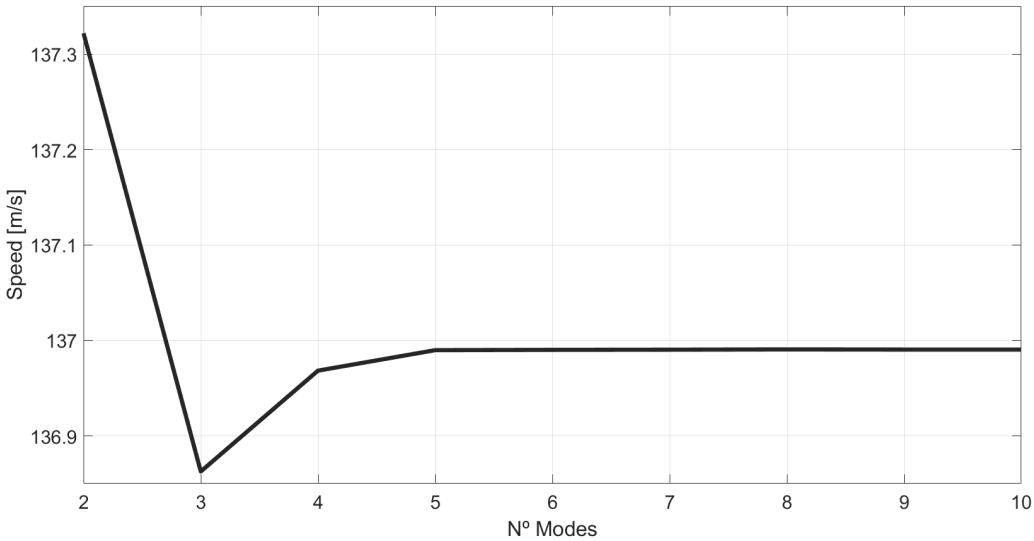


Figure 4: Flutter speed vs Nº of proper orthogonal modes employed

The principal conclusion it can be obtained from the results is that the flutter is principally affected by the motion of the second mode, since only considering the first 2 modes already gets a speed within a 1% error. If just 1 mode was considered, no flutter occurs (at least in the wide range of 0 to 300 m/s).

4 Sensitivity to parameters modifications

Objective: Obtain the sensitivity of the flutter speed with respect to some parameters

In this section, 4 different parameters will be modified in the same code as the last section (considered now as inputs) within the 2% range in order to obtain its influence in the flutter speed, being the following:

- **Structural bending stiffness** (EJ)
- **Structural torsional stiffness** (GJ)
- **Center of Gravity (CG) position**(xt_n)
- **Lift slope coefficient** ($C_{L\alpha}$)

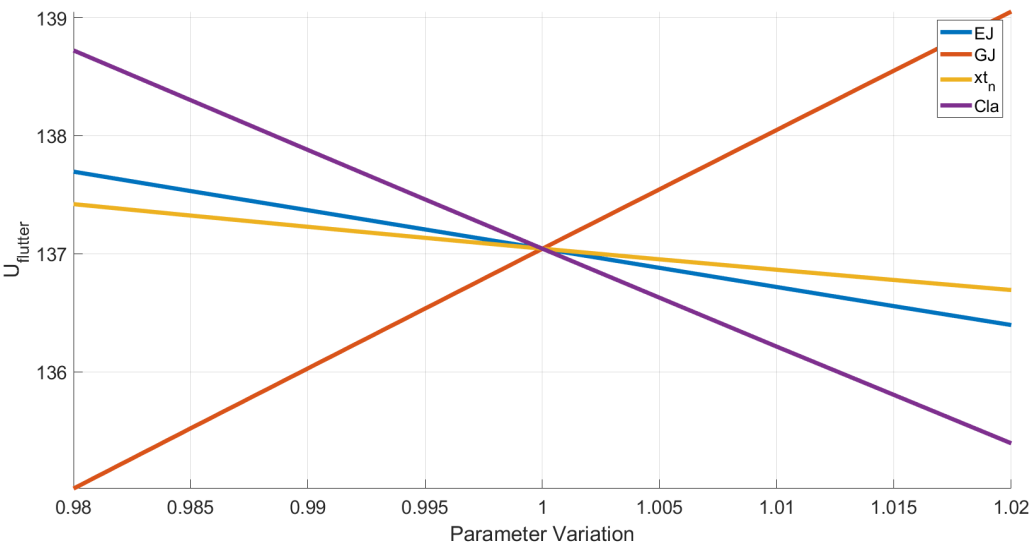


Figure 5: Flutter speed vs Variation of parameters

The approximate derivatives of the different parameters is:

Parameter	**Symbol** (p)	**Sensitivity** $\left(\frac{\partial U_f}{\partial \hat{p}}\right)$
Bending Stiffness	EI	-32
Torsional Stiffness	GJ	101
CG Position	xt_n	-18
Lift Slope	$C_{L\alpha}$	-83

Table 2: Sensitivity analysis results: Partial derivatives of Flutter Speed with respect to normalized parameters.

5 Addition of a Tip Weight

Objective: Analysis of the flutter speed with the addition of a wing tip mass

In this section, the addition of a tip wing weight will be analyzed, expecting that in some position chordwise the flutter speed is increased by some prescribed amount. This is a measure that could be considered if flutter was an important limitation in the correct work of the wing.

For this section on, it has been used the author's own code, using global Ritz functions for the modes and a P-K method for flutter solving. The additional terms coming from the wing tip mass in the virtual work equations are the following:

$$\delta W_{i,aileron} = \delta W_i + \delta\omega(L) m_t \ddot{\omega}(L) + \delta\theta(L) J_{t,CE} \ddot{\theta}(L) - \delta\omega(L) m_t s_t \ddot{\theta}(L) - \delta\theta(L) m_t s_t \ddot{\omega} \tag{2}$$

Below, the plotting of the flutter speed variation with chordwise position of the mass is showed:

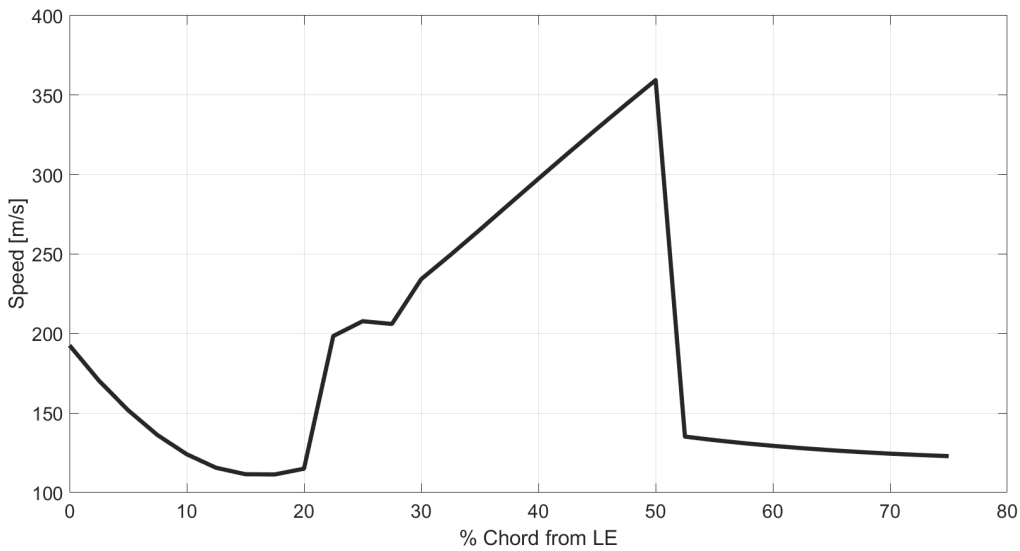


Figure 6: Flutter speed vs Wing tip mass chordwise position

Case	**Tip Weight CG Position**	**Flutter Speed (U_f) [m/s]**
a)	Coincident with Elastic Axis	250
b)	5% of the chord (0.05c)	151
c)	50% of the chord (0.50c)	360

Table 3: Effect of the tip weight CG position on the flutter speed.

If the wing tip mass is placed in the 38.5 % of the chord, an increase in 15% of flutter speed with respect to configuration a is obtained.

6 Mach number correction

Objective: Analyze the sensitivity of flutter speed with Mach number

The correction used is the Prandtl-Glauert, that introduces the Mach number variation is in the lift coefficient, so the impact will be similar as the studied in the section 4.

$$Cl_\alpha = \frac{Cl_\alpha(M=0)}{\sqrt{1-M^2}} \tag{3}$$

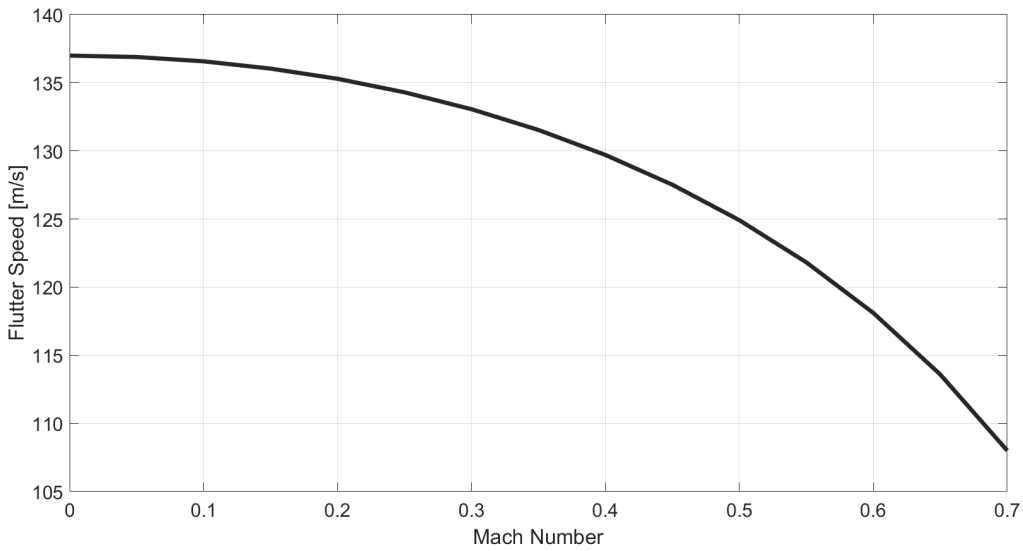


Figure 7: Prandtl-Glauert correction

A parabolic decrease of flutter speed with Mach is observed with the Prandtl-Glauert correction.

7 Aileron addition

Objective: Study the influence of the addition of a simple rigid aleiron in the Goland Wing model

Using the geometry provided in the introduction slides of the course, the aileron was defined in the code by placing equivalent forces in it's middle span point. For the aerodynamic forces the integration of the shape functions have been done, so they will be considered spanwise and not concentrated in a point.

The additional terms in the virtual work equation are:

$$\begin{aligned}\delta W_{i,aileron} = \delta W_i &+ \delta\beta\left[I_\beta\ddot{\beta} + \left(I_\beta + m_A s_\beta \frac{c_A}{2}\right)\ddot{\theta} - m_A\frac{c_A}{2}\ddot{\omega}\right] \\ &+ \delta\omega\left(m_A\ddot{\omega} - m_A\frac{c_A}{2}\ddot{\beta}\right) \\ &+ \delta\theta\left[(I_\beta + m_A s_\beta)\ddot{\theta} + \left(I_\beta + m_A s_\beta\frac{c_A}{2}\right)\ddot{\beta}\right]\end{aligned} \tag{4}$$

Also the modal shapes integrals needed for the aerodynamic virtual work are computed. The new flutter speed obtained is:

$$V_{flutter,aileron} = 117 \text{ m/s} \tag{5}$$

For the computation of the control reversal, the following procedure has been taken, considering that δ is the control surface angle ordered by the pilot:

$$A = \begin{bmatrix} K_{\omega\omega} & 0 & 0 \\ 0 & K_{\theta\theta} & 0 \\ 0 & 0 & K_{\beta\beta} \end{bmatrix} - U^2 \begin{bmatrix} 0 & K^A_{\omega\theta} & K^A_{\omega\beta} \\ 0 & K^A_{\theta\theta} & K^A_{\theta\beta} \\ 0 & K^A_{\beta\theta} & K^A_{\beta\beta} \end{bmatrix} \tag{6}$$

$$A\begin{Bmatrix} w \\ \theta \\ \beta \end{Bmatrix} = \begin{bmatrix} 0 \\ 0 \\ K_{\beta\beta} \end{bmatrix}\delta \tag{7}$$

$$\begin{Bmatrix} \omega \\ \theta \\ \beta \end{Bmatrix} = \begin{bmatrix} \omega_\delta \\ \theta\delta \\ \beta\delta \end{bmatrix}\delta \tag{8}$$

$$L = \int L_\theta \theta dx + \int L_\beta \beta dx \rightarrow \partial L/\partial\delta = 0 \tag{9}$$

When this condition is satisfied, the control reversal takes place. For this Goland wing model, the control reversal speed is:

$$V_{CR} = 158 \text{ m/s} \tag{10}$$

The divergence speed has also been obtained, with the condition $K - K^A = 0$:

$$V_{DV} = 223 \text{ m/s} \tag{11}$$